

Tagurit

Data Foraging in Disconnected Environments

Checkpoint 1

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The Setting

Simple premise:

- Drone or robot in the wild wants to send image data to cloudlet for computation, coordination, etc.
 - ie. Search and rescue, looking for a person

This assumes we can rely on the network



The Problem



We cannot rely on the network.
(Murphy always strikes)

Motivation

- Cannot guarantee network connectivity in the field
 - Signals can be jammed by the enemy
 - Might enter an area with little to no network infrastructure
 - Physical obstructions, bad weather
- ... But we have a lot of data to send → Becomes Stale

Need some way to prioritize what to send on Reconnection



Proposal

- Ingest image data from camera
- Good connection
 - Tag interesting image data with priority → Send to Cloudlet
- Disconnection
 - Store prioritized image data (Hoard)
- Reconnection
 - Concurrently send Stored Bank + Live data



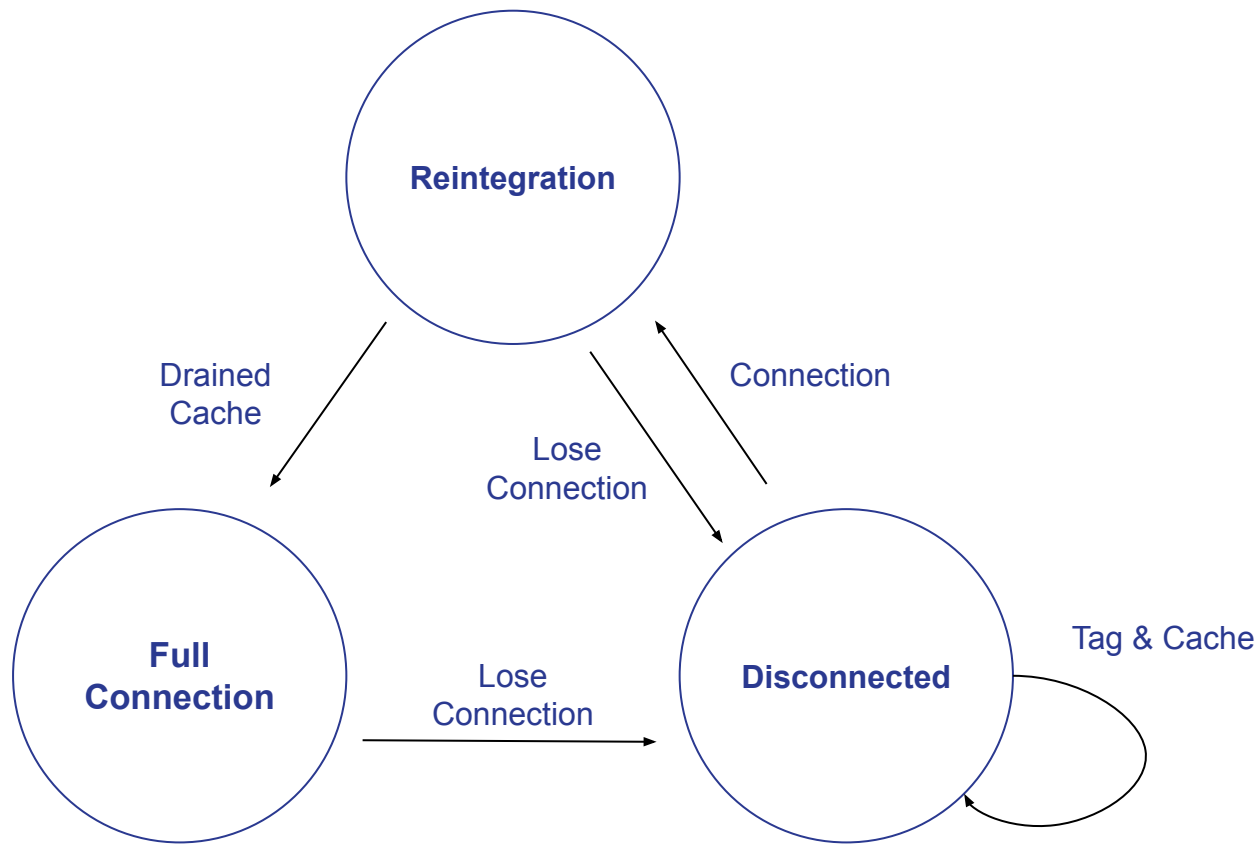
Proposed Architecture

Let's visualize as a simple DFA

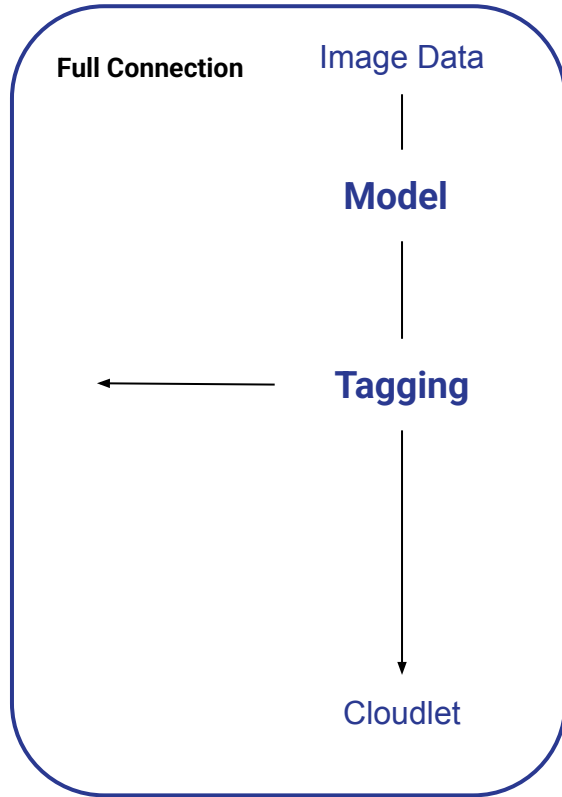
(technically NFA for those passionate about automata)



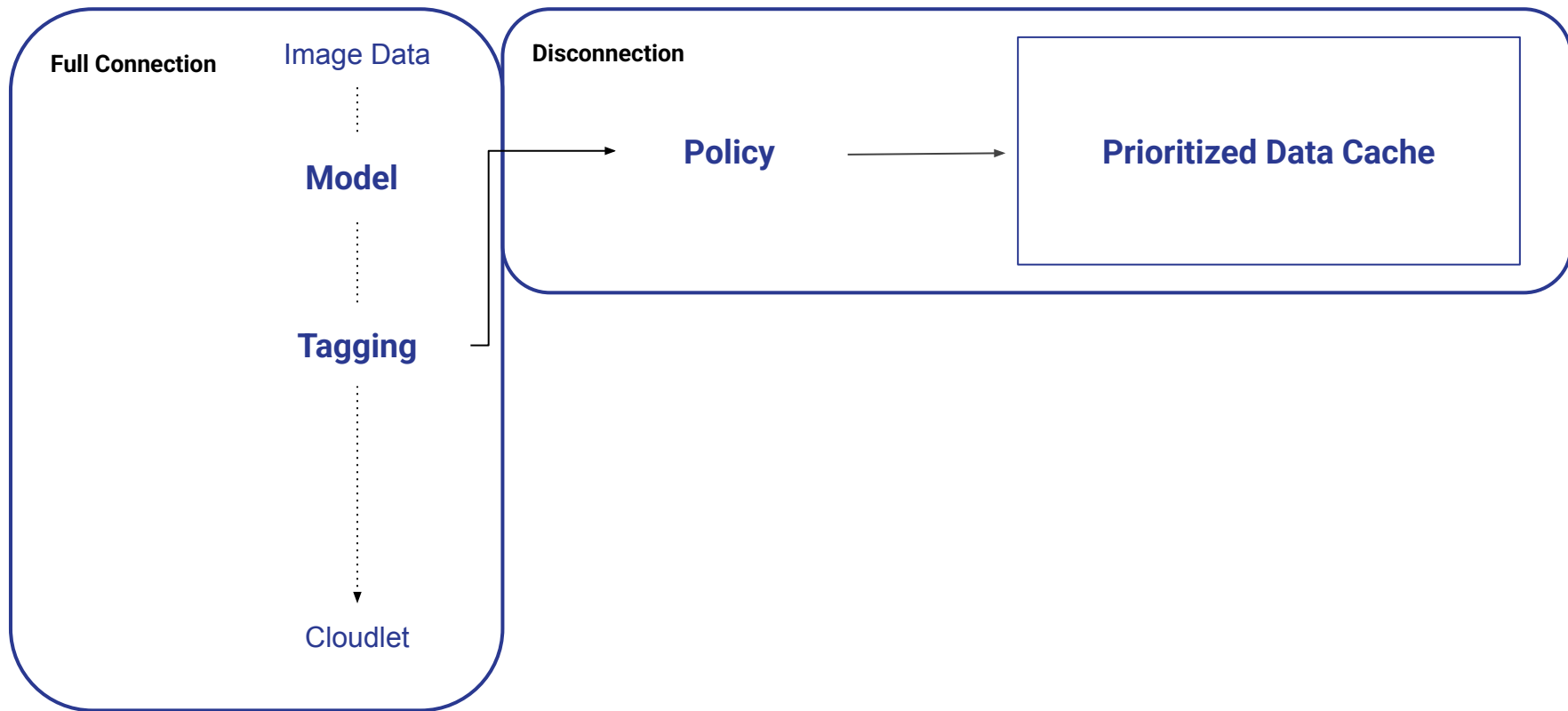
Proposed Architecture



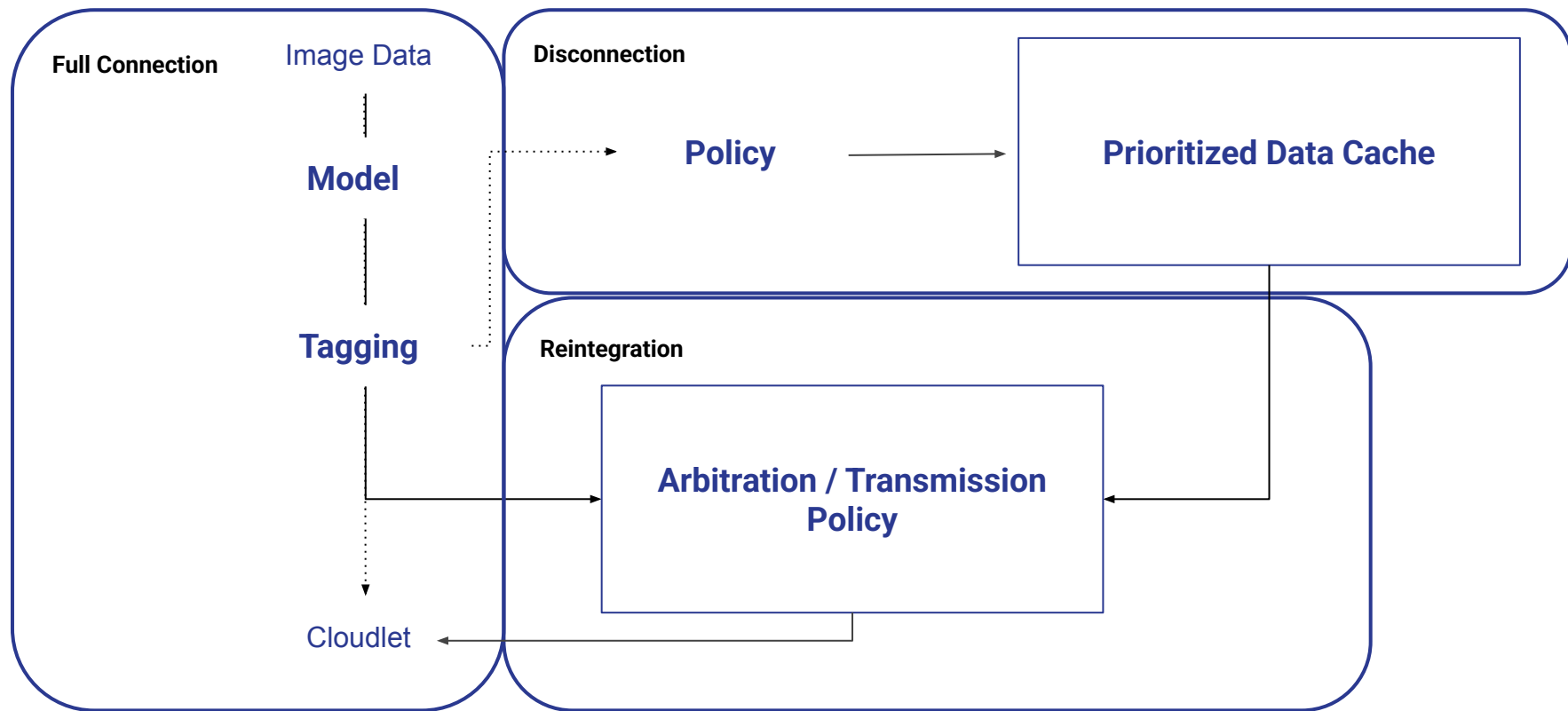
Proposed Architecture



Proposed Architecture



Proposed Architecture



Initial Prototype Stack

Edge Compute: NVIDIA Jetson Orin

Cloudlet: Remote compute node for heavier image processing

Onboard Model: YOLO

- Pretrained on COCO for initial object detection / priority tagging

Data Foraging Layer: Custom cache + queue architecture

- Stored prioritized image cache
- Live image queue

Communication Middleware: Gabriel

- Open-source CMU SatyaLab edge - cloud framework
- Client on edge device, server on cloudlet
- gRPC/TCP transport

Division of Work

Elias - Onboard Model & Tagging Policy

- Lightweight YOLO inference on edge device
- Define “interesting” image criteria
- Generate per-frame priority score
- Evaluate tagging accuracy / recall
- Define handoff format: image + priority + metadata

Yash - Data Foraging & Communication System

- Connectivity state management
 - Connected / Disconnected / Reconnecting
- Stored data queue management
- Live vs stored data arbitration
- Reconnection transmission scheduling
- Gabriel client/server integration
- Network Performance measurement

System Boundary: Elias outputs a scored frame → Yash decides whether to retain it, when to transmit it, and sends it through Gabriel to the cloudlet.

Project Management

- We have met with our mentors and established project scope/requirements /expectations
- Weekly meetings with Mihir and Thomas established
 - Fridays at 10:00 A.M.
- Elias and Yash will meet weekly to discuss progress on separate components
 - Will meet more often for integration based implementation
- Emergency meetings we will play by ear



Project Management

Early progress:

- Code repository has been scaffolded
 - Stalk us here: github.com/ewzheng/tagurit
- Arranged access to cloudlet and edge compute
 - Tom has provided us with access to a cloudlet for the project
 - We will be using a Jetson Orin NX 8GB to simulate a drone/robot
- Initial Familiarity with YOLO and Gabriel Established



Project Management



Pre-Checkpoint

- Organize team
- Discuss project scope

Project Management



Pre-Checkpoint

- Organize team
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Checkpoint 1

- Define goals
- Design architecture

Project Management



Pre-Checkpoint

- Organize team
- Discuss project scope

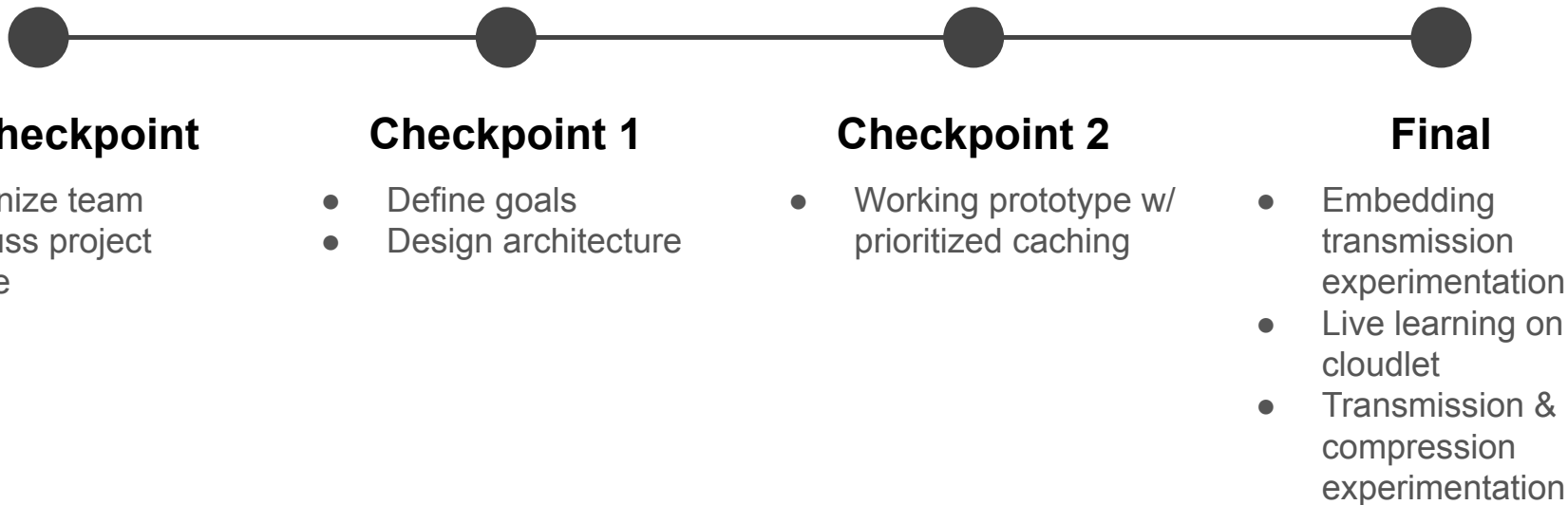
Checkpoint 1

- Define goals
- Design architecture

Checkpoint 2

- Working prototype w/ prioritized caching

Project Management



Fallback Strategies

Minimum Viable Product

- Stream all data during full connection
- Cache all data during disconnection
- Interleave cached and live data during reconnection

Less Minimal Product

- Develop a tagging heuristic
- Develop cache inclusion policy



Goals

- Design and implement a data tagging and caching system
 - The system must be lightweight and run on edge compute
- Ideally experiment with more sophisticated methods
 - Different caching and transmission policies
 - Investigate caching embeddings rather than raw data
- Demo on Jetson Orin NX communicating w/ cloudlet

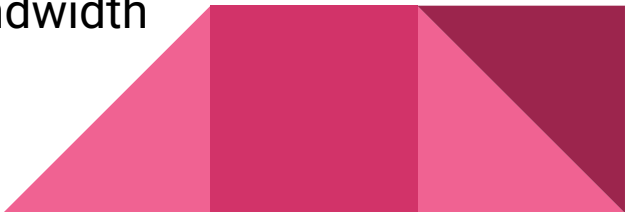


Stretch goals

Machine Learning

- Embedding transmission and lookup implementation
- Live learning (ie. cloudlet retraining model live on new class)

Adaptive connectivity aware transmission

- Detect weak or bandwidth-constrained connections
 - Dynamically adjust transmission behavior based on available network capacity
 - Adaptive image compression based on available bandwidth
 - Replacing Gabriel with a custom transport scheme
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Thank you!

Questions?
